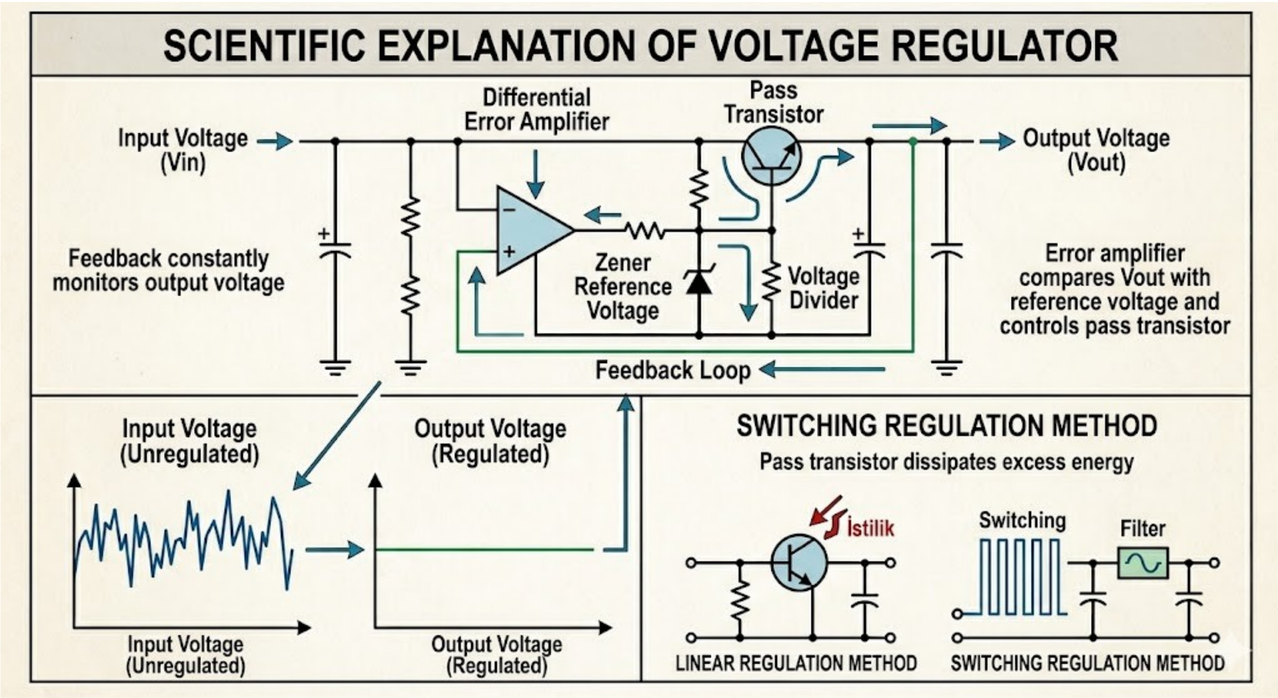
Physics Guide: COMPREHENSİVE TECHNİCAL REPORT: Voltage Regulator

Classification: TECHNICAL GUIDE/ EDUCATIONAL

Xumar Bağıyeva, Aysel İsmayılova, Rəşadə Zülfüqarlı, Yasin Quluzade,Ahmed Abdullayev

Verified by: Physics Teacher Azerbaijan Telman Askeraliyev (Fizika Muellimi, Azerbaijan, Baku)



Voltage regulators

A voltage regulator keeps the output voltage stable,

even if the input or load changes.

Topic 1

Voltage

regulation

Topic 2

Linear series

Topic 3

Linear shunt

Topic 4

Switching

• Used in power supplies, phone chargers, and embedded systems.

• Goal: Vout stays constant regardless of Vin or load changes.

Topic 1

Voltage regulation

How well does the regulator keep Vout stable?

Regulation = (V_no-load - V_full-load) / V_full-load x 100%

• Ideal regulator: 0% — Vout never changes.

• Real regulators have a small change — lower % is better.

• Two main problems: load regulation (current changes) and line regulation (input voltage changes).

Topic 2

Basic linear series regulator

A transistor in series controls the output voltage.

Position

Series with load

Control element

Efficiency

Transistor (BJT)

Low (~50%)

• The transistor acts like a variable resistor — it drops extra voltage.

• Simple and clean output — good for low-noise applications.

• Disadvantage: wasted energy becomes heat.

Topic 3

Basic linear shunt regulator

A transistor in parallel (shunt) diverts extra current away.

Position

Parallel with load

Control element

Efficiency

Transistor / Zener

Very low

• Excess current is sent through the shunt transistor, not the load.

• Very simple — Zener diode is the basic version of this.

• Disadvantage: wastes power even when load is small.

Topic 4

Basic switching regulator

Turns the transistor ON/OFF rapidly to control output voltage.

Method

PWM switching

Efficiency

Complexity

High (85-95%)

Higher

• Uses an inductor and capacitor to smooth the switched output.

• Can step up (boost), step down (buck), or invert voltage.

• Used in laptops, phone chargers — anywhere efficiency matters.

Buck (step-down)

Boost (step-up)

Buck-boost (both